

Data Dictionary: Borrow-Box

This document provides a comprehensive Data Dictionary for the system. It maps out each database entity (model), along with all its attributes, data types, constraints, and descriptions.

1. User (Django Built-In auth.User)

This represents the authentication table for users logging into the system, functioning as both renters and potential owners.

Field Name	Data Type	Length	Constraints / Keys	Default	Description
id	Integer	-	Primary Key, Auto-increment	-	Unique identifier for the user.
username	Varchar	150	Unique, Not Null	-	Username used for login.
password	Varchar	128	Not Null	-	Hashed password for security.
email	Varchar	254	Not Null	-	Email address of the user.
first_name	Varchar	150	-	blank	First name of the user.
last_name	Varchar	150	-	blank	Last name of the user.
is_superuser	Boolean	-	Not Null	False	Flag denoting admin privileges.

2. Profile (app_profile)

An extension of the User model, dedicated to holding additional user details like contact information and evaluating seller verification workflows.

Field Name	Data Type	Length	Constraints / Keys	Default	Description
id	Integer	-	Primary Key, Auto-increment	-	Unique profile identifier.
user_id	Integer	-	Foreign Key (User), Unique (OneToOne)	-	Associates the profile exclusively with one user.
is_seller	Boolean	-	Not Null	False	Boolean flag indicating actual seller privileges.
seller_status	Varchar	20	Choices: none, pending, approved, rejected	'none'	Workflow state for the seller application process.
phone	Varchar	15	Nullable, Blank Allowed	Null	Optional phone number of the user.
address	Text	-	Nullable, Blank Allowed	Null	User's detailed physical address.

3. Category (app_category)

A lookup table holding different groupings or classes of rental items.

Field Name	Data Type	Length	Constraints / Keys	Default	Description
id	Integer	-	Primary Key, Auto-increment	-	Unique category identifier.
name	Varchar	100	Unique, Not Null	-	Name of the item category.

4. Item (app_item)

Holds details for individual rental items uploaded to the platform by owners.

Field Name	Data Type	Length	Constraints / Keys	Default	Description
id	Integer	-	Primary Key, Auto-increment	-	Unique item identifier.
owner_id	Integer	-	Foreign Key (User), Not Null, CASCADE delete	-	References the user listing the item.
category_id	Integer	-	Foreign Key (Category), Nullable, SET_NULL	Null	References item's category. Enables related items.
title	Varchar	200	Not Null	-	Title or name of the rental item.
description	Text	-	Not Null	-	Extensive description about the item's condition.
price_per_day	Decimal	(8, 2)	Max Digits: 8, Decimal Places: 2, Not Null	-	Cost per day for renting the item.
location	Varchar	200	Not Null	-	Physical pickup/city location of the item.
image	Varchar	100	Upload path mapping (items/), Not Null	-	Path to the image file showcasing the item.

5. Booking (app_booking)

Serves as the central transaction/order table joining a user (renter) to an item for a specified timespan.

Field Name	Data Type	Length	Constraints / Keys	Default	Description
id	Integer	-	Primary Key, Auto-increment	-	Unique order/booking identifier.
item_id	Integer	-	Foreign Key (Item), Not Null, CASCADE delete	-	The specific item being rented.
user_id	Integer	-	Foreign Key (User), Not Null, CASCADE delete	-	The user acting as the renter.
start_date	Date	-	Not Null	-	The start date of the rental period.
end_date	Date	-	Not Null	-	The end date of the rental period.
total_price	Decimal	(10, 2)	Max Digits: 10, Decimal Places: 2, Not Null	-	Total sum of the booking (price_per_day * days).
status	Varchar	20	Choices: pending, confirmed, expired, cancelled	'pending'	Transacted state of the checkout process.
created_at	DateTime	-	Auto Add Now, Not Null	Timestamp	Absolute timestamp when the user clicked 'add to cart'.
expires_at	DateTime	-	Nullable, Blank Allowed	Null	System checkout timeout date (created_at + 5 Mins).

6. Contact (app_contact)

A standalone entity capturing feedback, queries, or messages from visitors and users on the frontend contact form.

Field Name	Data Type	Length	Constraints / Keys	Default	Description
id	Integer	-	Primary Key, Auto-increment	-	Unique contact request identifier.
first_name	Varchar	100	Not Null	-	Requester's first name.
last_name	Varchar	100	Not Null	-	Requester's last name.
email	Varchar	254	Not Null, Email Format	-	Requester's contact email.
phone	Varchar	15	Not Null	-	Requester's contact phone number.
subject	Varchar	200	Not Null	-	Headline/Topic of the message.
message	Text	-	Not Null	-	The actual body text of the message.